Environmental Data Science: Homework #7
Linear Classification
Due: Before Class on 10/14/26

Linear classification in python – EPA AQS Data

Using the EPA AQS data make a scatter plot of O_3 vs NO_2 concentrations and O_3 vs $PM_{2.5}$ concentrations.

1. Using these data as input, train a ridge classifier to predict if the measurements come from summer or winter.
2. What cost function are you using?
3. What is the optimal ridge penalty parameter based on your training data?
4. What is the optimal ridge penalty parameter on your validation data?
5. What is your final model error? (Be sure to report the most honest error assessment).
6. If you redo your train/validation splits, do you get the same answers?
7. Compare the ridge classifier with a logistic regression model. Which one performs the best?